

A certified continuation machine for Mahler measure identities at CM points, with seven new proofs of conjectures of Samart

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Declaration on the use of AI tools

The research reported in this article — including the computational exploration, the discovery of the proof strategy, the machine-certified verifications, and the preparation of the manuscript — was carried out by the author with the assistance of the AI system *Kimi* (Moonshot AI). All mathematical content, including every proof and every certified computation, has been checked and verified by the author, who takes full responsibility for the correctness and integrity of the article. All certification scripts are available for independent verification (see Section 2.5).

Abstract

We axiomatize the differential-comparison continuation method of [5] into a general machine for proving Mahler measure identities at CM points: a modular parametrization with rigorous tail bounds, a single-valued holomorphic Mahler differential on the complement of the critical image, a propagation lemma that replaces all monodromy and universal-cover arguments, interval-arithmetic path certification, and an exact three-track CM evaluation. Applied to Samart’s family $n_4(s) = 4m(x^4 + y^4 + z^4 + 1 + s^{1/4}xyz)$, the machine proves seven further entries of his 2015 table [2, Table 6]: the conjugate pairs at $s = 8292456 \pm 3132675\sqrt{7}$ (whose two identities are proved separately, not merely as a sum), $s = 3656 \pm 2600\sqrt{2}$, $s = -144$, $s = 143208 + 101574\sqrt{2}$, and the value $s = 1207368 + 853632\sqrt{2} + 697680\sqrt{3} + 493272\sqrt{6}$ attached to the class-number-two field $\mathbb{Q}(\sqrt{-6})$ — the first Mahler measure identity proved at a CM point of class number two, where the lattice sums involve two newforms and a genus character. New methodological ingredients include: the Fricke ratio as the only non-vacuous level/root-number discriminator (the “functional-equation residue” test is vacuous for root number +1 at every level); a genuine twist-level trap ($g_{16} \otimes \chi_8$ has level 64, not 32, and the wrong level silently scales the L -value); a genus-character decomposition of the lattice sums in class number two; and the discovery that Samart’s applicability boundary $\{\text{Im}\tau = 1/\sqrt{2}\}$ is not the topological boundary of the continuation region V_4 . Every identity is certified by an exact algebraic track (rational arithmetic only), rigorous interval locks (half-widths down to 9.1×10^{-53}), and independent 50–60-digit numerical cross-checks; all certification scripts are public. Finally we score the ten remaining open entries of Table 6 against the machine’s five stations: five are within current reach (only mechanical work remains), and five are blocked by genuine obstructions — a two-dimensional critical image or a wrong-sheet Eisenstein–Kronecker series — which we analyze precisely.

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1 Introduction

1.1 The question

Let $m(P)$ denote the logarithmic Mahler measure of a Laurent polynomial and let

$$n_4(s) := 4 m(x^4 + y^4 + z^4 + 1 + s^{1/4}xyz), \quad s \in \mathbb{C},$$

the value being independent of the choice of the fourth root (Section 2). In [1, 2] Samart expressed $n_4(s)$, for s in the image of a modular parametrization $s_4(\tau)$, as an Eisenstein–Kronecker (EK) series, proved the identity for pure imaginary τ with $\text{Im}\tau \geq 1/\sqrt{2}$, and conjectured on numerical evidence the twenty-nine L -value evaluations of his Table 6 [2]: each attaches to a CM value of s a rational linear combination of L -values of CM weight-three newforms and of quadratic Dirichlet characters.

The table is now largely, but not entirely, resolved. Eleven entries were proved previously: the exterior values $s = 256$ (Rogers [8]), $s = 648, 2304, 20736, 614656$ (Samart [1]), $s = 26856 \pm 15300\sqrt{3}$ (Samart’s own Theorem 3.1 [2]), the outside-the-critical-locus value $s = -3969$ (Fei [4]), and $s = -1024, -12288, -82944$ (Guo–Peng–Qin [11]). One entry is *refuted*: the conjecture $n_4(81) = 40M_7$ fails numerically as a statement about the true Mahler measure, the conjectured value being the evaluation of the EK series at the attached CM point on a wrong sheet [6]. He–Ye [3] resolved the companion family n_3 completely; the n_2 family was completed in [5, 10, 11]. Fei [4] proved, for all rational singular moduli in twenty-three families, identities at the level of the real part of the *holomorphic* Mahler measure $\text{Re}\tilde{m}$; at parameters inside the critical locus the passage from $\text{Re}\tilde{m}$ to the genuine Mahler measure is not automatic — a point stressed already by Rodríguez Villegas [9] — and it is precisely this passage that the continuation machine of [5] and of the present paper performs.

The purpose of this paper is threefold:

- (i) to axiomatize the differential-comparison continuation method of [5] into a general, family-independent machine (Section 2), with an explicit certifier architecture whose trusted base consists of pure mathematics, documented interval-arithmetic semantics, and public source code (Section 2.5);
- (ii) to apply the machine to seven further entries of Samart’s Table 6 (Theorems A–F below), covering all four regimes that occur: deep pure-imaginary interior points (A, C, D, E), a non-pure-imaginary interior point (B), the first class-number-two CM point (E), and a point on Samart’s applicability boundary that turns out to be a strict interior point of the continuation region (F);
- (iii) to take stock: to score the ten remaining open entries of Table 6 against the machine’s checklist (§1.3) and to describe precisely where and why the method stops (Section 8).

1.2 Results of this paper

Throughout, $d_k := L'(\chi_{-k}, -1)$ for the odd quadratic Dirichlet character χ_{-k} of conductor k , and M_k or $M_k^{(i)}$ denotes $L'(g_k, 0)$, resp. $L'(g_k^{(i)}, 0)$, for the indicated weight-3 CM newform(s); a superscript *tw* denotes the L' -value of an explicitly identified quadratic twist.

Theorem 1.1 (Theorem A: the $\sqrt{7}$ pair). *Let $M_7 := L'(g_7, 0)$ with $g_7(\tau) = \eta(\tau)^3\eta(7\tau)^3$ (LMFDB 7.3.b.a) and $M_7^{\text{tw}} := L'(g_7 \otimes \chi_{-4}, 0)$. Then*

$$n_4(8292456 + 3132675\sqrt{7}) = \frac{5}{28}(4M_7^{\text{tw}} + 224M_7 + 32d_4 + 7d_7), \quad (1)$$

$$n_4(8292456 - 3132675\sqrt{7}) = \frac{5}{14}(4M_7^{\text{tw}} - 224M_7 + 32d_4 - 7d_7). \quad (2)$$

Theorem 1.2 (Theorem B). *With $M_{12} := L'(g_{12}, 0)$, $g_{12}(\tau) = \eta(2\tau)^3\eta(6\tau)^3$ (LMFDB 12.3.c.a),*

$$n_4(-144) = \frac{10}{3}(4M_{12} + d_3).$$

Theorem 1.3 (Theorem C). *With $M_8 := L'(g_8, 0)$, $g_8(\tau) = \eta(\tau)^2\eta(2\tau)\eta(4\tau)\eta(8\tau)^2 \in S_3(\Gamma_0(8), \chi_{-8})$, and $M_8^{\text{tw}} := L'(g_8 \otimes \chi_8, 0)$, where $\chi_8 = (2/\cdot)$ is the even primitive character of conductor 8,*

$$n_4(3656 + 2600\sqrt{2}) = \frac{5}{8}(4M_8^{\text{tw}} + 28M_8 + 4d_4 + d_8).$$

Theorem 1.4 (Theorem D). *With $M_{16} := L'(g_{16}, 0)$, $g_{16}(\tau) = \eta(4\tau)^6 \in S_3(\Gamma_0(16), \chi_{-4})$, and $M_{16}^{\text{tw}} := L'(g_{16} \otimes \chi_8, 0)$ (level 64 — see §5.4),*

$$n_4(143208 + 101574\sqrt{2}) = \frac{5}{16}(4M_{16}^{\text{tw}} + 20M_{16} + 9d_4 + 4d_8).$$

Theorem 1.5 (Theorem E: class number two). *Let $K = \mathbb{Q}(\sqrt{-6})$ (class number 2), and let $g_{24}^{(1)}, g_{24}^{(2)}$ be the two newforms of $S_3(\Gamma_0(24), \chi_{-24})$, theta series of the two grössencharacters of K of type (2, 0) (LMFDB 24.3.h.a and 24.3.h.b). With $M_{24}^{(i)} := L'(g_{24}^{(i)}, 0)$ and $M_{24}^{(i), \text{tw}} := L'(g_{24}^{(i)} \otimes \chi_{-8}, 0)$ (level 96),*

$$\begin{aligned} n_4(1207368 + 853632\sqrt{2} + 697680\sqrt{3} + 493272\sqrt{6}) \\ = \frac{5}{48} \left(4M_{24}^{(1), \text{tw}} + 4M_{24}^{(2), \text{tw}} + 28M_{24}^{(1)} + 12M_{24}^{(2)} + 28d_3 + 24d_4 + 8d_8 + d_{24} \right). \end{aligned}$$

Theorem 1.6 (Theorem F: a boundary point that is interior). *With the notation of Theorem 1.3,*

$$n_4(3656 - 2600\sqrt{2}) = \frac{5}{4}(4M_8^{\text{tw}} - 28M_8 + 4d_4 - d_8).$$

The s -value is the conjugate of Theorem 1.3, but the attached CM point $\tau_4 = (1 + \sqrt{-2})/2$ lies exactly on Samart's applicability boundary $\text{Im}\tau = 1/\sqrt{2}$, where his proof does not apply; the identity is proved by the continuation machine (Section 7), which shows that τ_4 is a strict interior point of the continuation region V_4 (certified margin 0.11590).

Theorem 1.7 (The machine, informal). *(Precise form: Theorem 2.10.) Let P_c be a tempered Laurent-polynomial family with a modular parametrization $c = c(\tau)$ and critical image $K \subset \mathbb{C}$, and let $E(\tau)$ be the Eisenstein–Kronecker expression for the holomorphic Mahler measure, valid on an open set $U \subset \mathbb{H}$. If (i) the period integral $\Omega(c)$ is single-valued and holomorphic on $\mathbb{C} \setminus K$, (ii) $c(\tau_0)$ is a CM parameter with $\tau_0 \in \overline{W}$, where W is the connected component of $\mathbb{H} \setminus c^{-1}(K)$ containing U , and (iii) the lattice sums underlying $E(\tau_0)$ evaluate exactly, then $m(P_{c(\tau_0)})$ equals the corresponding L -value expression; the inclusion $\tau_0 \in \overline{W}$ is certifiable by rigorous interval arithmetic.*

All seven identities are machine-certified. Concretely, each proof consists of an exact algebraic track (rational/algebraic-integer arithmetic only, no floating point), rigorous interval locks of the CM values and of the final identities (half-widths between 9.1×10^{-53} and 2.3×10^{-51}), and independent 50–60-digit numerical cross-checks; the scripts are listed in Section 2.5 and are publicly available (see the Data Availability statement). Two representative reference values:

$$\begin{aligned} M_8 &= 0.1353184954231269106150060654595395883532\dots, \\ M_8^{\text{tw}} &= 1.549084847095611486271658159444376281785\dots \quad (\text{level } 32, w = +1), \end{aligned}$$

computed from the functional equations and cross-checked against direct lattice-sum evaluations; further values are collected in the application sections.

1.3 The ten remaining entries

Table 6 of [2] has twenty-nine entries: eighteen are now proved (eleven previously known, plus the seven of Theorems A–F), one is refuted ($s = 81$ [6]), and ten remain open. The machine reduces a proof of each entry to five *stations* (Section 2 for definitions):

- S1** *Parametrization*: a CM point τ_0 with $s_4(\tau_0) = s$ identified, at least numerically;
- S2** *Critical image*: the position of $c(\tau_0)$ relative to the astroid disc Ast and the cross $[-4, 4] \cup i[-4, 4]$;
- S3** *Interior-or-boundary*: membership of τ_0 in $\overline{W}$, certified by a **cert2** path, or direct applicability of Samart’s theorem;
- S4** *cert0 algebraicity*: an exact algebraic lock of the value $s_4(\tau_0)$;
- S5** *P1 assemblability*: an exact CM evaluation of $\text{EK}_4(\tau_0)$ in the relevant imaginary quadratic field.

Table 1 scores the ten open entries against these stations.

The conjectured right-hand sides are Samart’s [2, Table 6]; for example, entry #1 is $n_4(143208 - 101574\sqrt{2}) \stackrel{?}{=} \frac{5}{8}(4M_{16}^{\text{tw}} - 20M_{16} - 9d_4 + 4d_8)$, the conjugate sign pattern of Theorem 1.4, and entries #2–#4 are the three remaining sign patterns of Theorem 1.5’s family. Five entries are *ready*: for #1 and #7 the target CM point even lies on the already certified path of Section 7 (legs $\{x = 1/2, 0.702 \leq y \leq 1\}$ and $\{y = 1, 0 \leq x \leq 1/2\}$), so only stations S4/S5 remain; #2, #6, #8 require a routine **cert2** adaptation and a new **cert0** pair lock. The five *blocked* entries split into two genuine obstructions, analyzed in Section 8: the two-dimensional critical image (#4, #5: the $s = 81$ mechanism of [6]), and the wrong-sheet phenomenon of the EK series below $\text{Im}\tau = 1/\sqrt{2}$ (#3, #9, #10, and again #5), where the conjectured identity concerns the true Mahler measure but the series side of the machine has no valid input.

Beyond Table 6, three further open directions are discussed in Section 8: Samart’s “three modular L -value” hypothesis $s = 16 + 1600\sqrt[3]{2} - 1280\sqrt[3]{4}$ [2], which has no attached CM point in the s_4 -image; the exterior entries, for which a corrected formula of Samart–Tao $\tilde{n}$ -type [7] would be needed; and the Boyd-lineage two-variable conjectures, where no CM evaluation is available at all.

Table 1: The ten open entries of Samart’s Table 6, scored against the five stations of the machine. “Outside/inside” refers to the astroid disc Ast; ζ abbreviates 1207368, $a = 853632$, $b = 697680$, $d = 493272$ in rows 2–4: $s = \zeta \pm a\sqrt{2} \pm b\sqrt{3} \pm d\sqrt{6}$ with the indicated signs.

#	s	τ_0	$\text{Im}\tau_0$	S2: $c(\tau_0)$	S3: position	S4/S5 status	verdict
1	143208 $101574\sqrt{2}$	$-(1+2i)/2$	1	outside ($ s > 256$)	Ast interior; τ_0 lies on the certified path of Section 7	S4 done (conjugate of Thm. D lock); S5 ready ($\mathbb{Q}(i)$, $h = 1$)	ready
2	ζ , signs $(+--)$	$(1+i\sqrt{6})/2$	1.225	outside ($ s > 256$)	Ast interior; short cert2 extension	S4 done (cert0_n4.p4.t3.py); S5 ready ($h = 2$, §6)	ready
3	ζ , signs $(-+-)$	$i\sqrt{6}/4$	0.612	outside ($s > 256$)	Ast <i>exterior</i> : EK series wrong- sheeted below $1/\sqrt{2}$	S4 done (same lock); S5 moot	blocked
4	ζ , signs $(--+)$	$(2+i\sqrt{6})/4$	0.612	inside ($s \approx -2.45$)	Ast blocked (two- dimensional image, open mapping)	S4 done (same lock); S5 moot	blocked
5	$(-19230385995\sqrt{5})/2$	$+(3+i\sqrt{15})/6$	0.646	inside ($s \approx -6.17$)	Ast blocked (two- dimensional image); also exte- rior	S4 open (disc. -15 , $h = 2$); S5 needs level-15 pair	blocked
6	$(-19230385995\sqrt{5})/2$	$-(1+i\sqrt{15})/2$	1.936	outside	Ast interior; cert2 feasible on $\text{Ret} = 1/2$	S4 open (pair lock with #5); S5 ready ($h = 2$ machinery)	ready
7	$-893952516096\sqrt{3}$	$+(3+i\sqrt{21})/6$	0.764	outside (functional ≈ 3.02)	Ast interior; τ_0 lies on the certified path of Section 7	S4 open (pair lock with #8); S5 ready (level 84, $h = 2$)	ready
8	$-893952516096\sqrt{3}$	$-(1+i\sqrt{21})/2$	2.291	outside	Ast interior; cert2 feasible	S4 open (same pair lock); S5 ready	ready
9	$347648256141926400\sqrt{6}$	$+i\sqrt{42}/42$	0.154	outside ($c > 4$ real)	Ast <i>exterior</i> : wrong sheet, deep below $1/\sqrt{2}$	S4 open (disc. -168 , $h = 4$); S5 moot	blocked
10	$347648256141926400\sqrt{6}$	$-i\sqrt{42}/14$	0.463	outside ($s \approx 995$, $c > 4$ real)	Ast <i>exterior</i> : wrong sheet below $1/\sqrt{2}$	S4 open; S5 moot	blocked

2 The machine

This section axiomatizes the method of [5, §3–§5] in family-independent language. The running example is the n_4 -family, but the only family-specific inputs are the parametrization, the critical image, and the anchor identity; we display them as separate *stations* S1–S5, matching the checklist of §1.3.

2.1 Families, parametrizations, critical images

Definition 2.1 (Tempered family). A *tempered family* is a Laurent-polynomial family $P_c(x_1, \dots, x_n) = P(x_1, \dots, x_n) + c$, $c \in \mathbb{C}$, such that $c \mapsto m(P_c)$ is continuous on all of $\mathbb{C}$ — for the families here this holds by the elementary dominated-convergence argument of [5, Lemma 2.1]. Its *critical image* is the closure $K \subset \mathbb{C}$ of the set of critical values of P on $\mathbb{T}^n$; on $\mathbb{C} \setminus K$ the polynomial P_c does not vanish on $\mathbb{T}^n$, the integrand $\log |P_c|$ is real-analytic in c , and $c \mapsto m(P_c)$ is real-analytic.

Two families serve as running examples.

Example 2.2 (The n_2 family [5]). $P + c$ with $P = (x + x^{-1})(y + y^{-1})(z + z^{-1})$ and $c = \sqrt{s_2(\tau)}$, $s_2(\tau) = q^{-1} \prod_{n \geq 1} (1 + q^{2n-1})^{24}$. The critical image in the c -plane is the slit $[-8, 8]$, and in the

s -plane the segment $[0, 64]$: one-dimensional.

Example 2.3 (The n_4 family). $P_c = (x^4 + y^4 + z^4 + 1)/(xyz) + c$ after clearing the monomial (Mahler measure is unchanged), with the natural *holomorphic* parametrization

$$c(\tau) = \left(\frac{\eta(2\tau)}{\eta(\tau)}\right)^6 (16A(\tau)^4 + A(\tau)^{-4}), \quad A(\tau) = \frac{\eta(\tau)\eta(4\tau)^2}{\eta(2\tau)^3}, \quad s_4(\tau) = c(\tau)^4. \quad (3)$$

Both c and s_4 are holomorphic on $\mathbb{H}$ — no branch choice is needed, in contrast to Example 2.2 — and s_4 has an integral q -expansion $s_4 = q^{-1}(1 + 104q + 4372q^2 + \dots) \in \mathbb{Z}((q))$ (exact integer check to order q^{30} in `cert0_s4_m144.py`). The critical image is *two-dimensional*: $P(\mathbb{T}^3)$ is the closed astroid disc

$$\text{Ast} = \{c \in \mathbb{C} : |\text{Rec}|^{2/3} + |\text{Imc}|^{2/3} \leq 4^{2/3}\}, \quad (4)$$

since P is a sum of four unit phasors with product 1 [6]. The branch cut of the holomorphic Mahler measure is the cross $\{c^4 \in (0, 256]\} = [-4, 4] \cup i[-4, 4]$, contained in Ast ; hence avoiding Ast automatically avoids every bad set, and we take $K = \text{Ast}$.

Remark 2.4. The dimension of the critical image is the single most important topological invariant of a family for this method. A one-dimensional slit does not separate the plane: every point of the slit lies in the closure of the good domain, and the propagation lemma below reaches boundary points by continuity (this is what makes the $k = 1$ proof of [5] possible). The two-dimensional astroid disc has interior: a CM parameter landing inside it (#4, #5, #10 of Table 1; the refuted $s = 81$) lies inside an open bad region, no path can approach it from the good domain (open mapping), and the machine stops. See Section 8.

Definition 2.5 (The good domain and the EK expression). For a holomorphic parametrization $c(\tau)$ set

$$V := \{\tau \in \mathbb{H} : c(\tau) \notin K\} \quad (\text{open, since } c \text{ is holomorphic}),$$

and let $F(\tau) := m(P_{c(\tau)})$ (times the family's normalization factor), continuous on all of $\mathbb{H}$ and real-analytic on V . An *EK datum* is a holomorphic function $E(\tau)$ on $\mathbb{H}$ with $G := 2\text{Re}E$ (or $\text{Im}E$, depending on normalization) real-analytic everywhere, such that $F = G$ on a nonempty open *anchor* set $U \subset V$. For the n_4 family (Example 2.3), with U_j as in (6) below,

$$E_4(\tau) := -i\left(2\pi\tau + \frac{10}{\pi^3}(U_1(\tau) - 2U_2(\tau))\right), \quad \text{EK}_4(\tau) := \text{Re}E_4(\tau) = \text{Im}\left[2\pi\tau + \frac{10}{\pi^3}(U_1 - 2U_2)\right], \quad (5)$$

holomorphic, resp. real-analytic, on all of $\mathbb{H}$; the double series

$$U_j(\tau) := \sum_{m \neq 0} \frac{1}{m} \sum_{n \in \mathbb{Z}} (jm\tau + n)^{-3} \quad (6)$$

converges absolutely, locally uniformly on $\mathbb{H}$ [5, §2.4]. The lattice-sum form used for all CM evaluations is

$$\text{EK}_4(\tau) = \frac{10 \text{Im}\tau}{\pi^3} (-T_1(\tau) + 4T_2(\tau)), \quad T_d(\tau) := \sum_{\lambda \in \Lambda_d(\tau)} \left[\frac{2\text{Re}\bar{\lambda}^2}{|\lambda|^6} + \frac{1}{|\lambda|^4} \right], \quad (7)$$

with $\Lambda_d(\tau) := \mathbb{Z} + \mathbb{Z}d\tau$; the series is absolutely convergent, and (7) follows from $w^{-3} - \bar{w}^{-3} = -2i(3u^2v - v^3)|w|^{-6}$ and $3u^2 - v^2 = 4u^2 - |w|^2$ exactly as in [5, Lemma 2.6].

2.2 The Mahler differential is single-valued

Proposition 2.6 (Mahler differential formula). *Let P_c be a tempered family and $K \subset \mathbb{C}$ its critical image. Then*

$$\Omega(c) := \int_{\mathbb{T}^n} \frac{d\mu}{P_c}$$

is single-valued and holomorphic on $\mathbb{C} \setminus K$, and as real 1-forms there,

$$dm(P_c) = \operatorname{Re}[-\Omega(c) dc]. \quad (8)$$

Proof. For any compact $K_0 \Subset \mathbb{C} \setminus K$ one has $|P_c| \geq \operatorname{dist}(K_0, K) > 0$ uniformly for $c \in K_0$ on $\mathbb{T}^n$, so differentiation under the integral sign is legitimate and gives $\Omega'(c) = \int_{\mathbb{T}^n} d\mu/P_c^2$; holomorphy and single-valuedness follow (the integral is defined over the *fixed* cycle $\mathbb{T}^n$). Writing $c = c_1 + ic_2$ and differentiating $m(P_c) = \int \log |P_c| d\mu$ in c_1, c_2 gives (8); see [5, Lemma 2.7]. $\square$

Remark 2.7. Proposition 2.6 is the point at which all monodromy and universal-cover arguments disappear from the method. The holomorphic Mahler measure $\tilde{m}$ of Villegas [9] is multivalued, but its multivaluedness lives in locally constant imaginary periods: every branch satisfies $\tilde{m}'(c) = -\Omega(c)$ on $\mathbb{C} \setminus K$. Comparing *differentials* rather than functions therefore never sees a branch choice, and the identity theorem below applies to a genuine single-valued holomorphic function.

2.3 The propagation theorem

Lemma 2.8 (Propagation lemma). *Let $U \subset \mathbb{C}$ be connected open, $Z \subset U$ relatively closed, and $H : U \rightarrow \mathbb{R}$ continuous and locally constant on $U \setminus Z$. Then H is constant on every connected component W of $U \setminus Z$; if $H|_W \equiv C$ then $H|_{\overline{W} \cap U} \equiv C$.*

Proof. A locally constant function is constant on each connected component. For the second assertion, let $p \in \overline{W} \cap U$ and choose $p_n \in W$ with $p_n \rightarrow p$; continuity gives $H(p) = \lim_n H(p_n) = C$. $\square$

Remark 2.9. The conclusion cannot be upgraded to “ H constant on U ”: the devil’s staircase is continuous, locally constant off the Cantor set, yet non-constant. Membership of the target point in $\overline{W}$ is a genuine topological input — supplied in our applications either by Samart’s proved region (Theorems A, C, D, E), or by an explicitly certified path (Theorems B, F), and *absent* for the blocked entries of Table 1.

Theorem 2.10 (Main theorem: differential-comparison continuation). *Let P_c be a tempered family with critical image K , let $c(\tau)$ be holomorphic on $\mathbb{H}$, set $V = \{\tau : c(\tau) \notin K\}$, and let $F(\tau) = m(P_{c(\tau)})$ (with the family normalization). Let $E(\tau)$ be holomorphic on $\mathbb{H}$, $G = 2\operatorname{Re}E$, and suppose $F = G$ on a nonempty open subset of a connected component W of V . Then*

$$F(\tau) = G(\tau) \quad \text{for every } \tau \in \overline{W} \cap \mathbb{H}. \quad (9)$$

Proof. On V both $\Omega(c(\tau))c'(\tau)$ and $E'(\tau)$ are holomorphic, so $\Psi(\tau) := \Omega(c(\tau))c'(\tau) + E'(\tau)$ is holomorphic on V . By the chain rule applied to (8), $dF = 2\operatorname{Re}[-\Omega(c(\tau))c'(\tau) d\tau]$ on V , and $dG = 2\operatorname{Re}[E'(\tau) d\tau]$ there; hence $d(F - G) = -2\operatorname{Re}[\Psi(\tau) d\tau]$. On the anchor open set $F - G = 0$, so $\operatorname{Re}[\Psi d\tau] = 0$ in all directions $d\tau$; testing $d\tau = 1$ and $d\tau = i$ gives $\Psi = 0$ there, and the identity theorem on the connected open set W gives $\Psi \equiv 0$ on W . Thus $H := F - G$ is locally constant on W , constant since W is connected, and zero because it vanishes on the anchor; H is continuous on all of $\mathbb{H}$ (Mahler continuity, Definition 2.1; G is real-analytic everywhere), so Lemma 2.8 with $U = \mathbb{H}$, $Z = c^{-1}(K)$ gives $H \equiv 0$ on $\overline{W} \cap \mathbb{H}$. $\square$

Remark 2.11 (The five stations). Instantiating Theorem 2.10 for a conjectured identity $n_4(s) = R$ at a CM parameter splits into the five stations of §1.3: S1 identifies τ_0 with $s_4(\tau_0) = s$; S2 locates $c(\tau_0)$ relative to Ast; S3 certifies $\tau_0 \in \overline{W}$ (or gets it for free from Samart’s proved region); S4 upgrades the numerical coincidence $s_4(\tau_0) = s$ to an exact algebraic lock (§2.5.3); S5 evaluates $\text{EK}_4(\tau_0)$ exactly as a rational linear combination of L -values (§2.5.4). Theorem 2.10 then yields $n_4(s_4(\tau_0)) = \text{EK}_4(\tau_0)$, and S4+S5 turn this into the identity $n_4(s) = R$.

2.4 The anchor identity for the n_4 family

The anchor input is Samart’s theorem, quoted with one repair.

Theorem 2.12 (Samart, [1, Prop. 2.1(iii)]). *For pure imaginary $\tau = iy$ with $y \geq 1/\sqrt{2}$, $n_4(s_4(\tau)) = \text{EK}_4(\tau)$.*

Lemma 2.13 (Composite continuation lemma: the anchor on $|s_4| > 256$). *Let $D \subset \mathbb{H}$ be open with $|s_4(\tau)| > 256$ for all $\tau \in D$. Then $n_4(s_4(\tau)) = \text{EK}_4(\tau)$ on D .*

Proof sketch with quoted inputs. Three quoted ingredients compose. (i) *Rogers’ evaluation* (Prop. 2.2 and Thm. 2.3 of [8]): the ${}_5F_4$ hypergeometric expression for n_4 converges for $|s| > 256$ ($\sum b - \sum a = 3/2 > 0$) and equals $n_4(s)$ there: both sides are real parts of holomorphic functions of s on $\mathbb{C} \setminus \text{Ast}$, equal on the real ray $s > 256$ by Rogers’ theorem, hence equal on the connected open set $\{|s| > 256\}$ by the identity theorem. (ii) *Rogers’ transformation* [8, Thm. 2.3]: $f_4(s_4(q)) = -\frac{1}{3}G(q) + \frac{2}{3}G(q^2)$ is an identity of functions analytic near $q = 0$, extending to $|q| < 1$ by the identity theorem. (iii) *Samart’s computation* [1, proof of Prop. 2.1(iii)]: the identification of the q -series in (ii) with the U -series expression EK_4 uses termwise integration and Fourier expansions whose convergence requires only $|q| < 1$. The single step of Samart’s proof that genuinely needs real q — the comparison $|s_4(q)| \geq s_4(e^{-\pi\sqrt{2}}) = 256$ by real-variable monotonicity, which locates his argument in the convergence region — is replaced here by the hypothesis $|s_4| > 256$ on D , certified by interval arithmetic in each application (Theorems 1.2 and 1.6). This repair was identified by a line-by-line check of [1, 2]; no other step of the proof uses the pure-imaginary hypothesis. $\square$

Lemma 2.13 upgrades Samart’s theorem from a statement on a half-line to an anchor on an open set: the anchor disk $D = \{|\text{Re}\tau| \leq 1/32, |\text{Im}\tau - 1| \leq 1/32\}$ carries the certified bound $|s_4| \geq 493$ (Section 4), so $F = G$ on D , and Theorem 2.10 propagates the identity through the whole component W of $V_4 := \{\tau : c(\tau) \notin \text{Ast}\}$ containing D . For the deep pure-imaginary points of Theorems A, C, D, E even this is unnecessary: Theorem 2.12 applies directly.

2.5 The certifier: cert0, cert2, verify_P1

The non-elementary computations are certified by interval arithmetic. The object entering the proofs is not a program but a *certificate*: finite data (parameter blocks with strict enclosures, or exact rational equalities) whose validity is checked by deterministic, finite, independently re-runnable computations. The trusted base consists of exactly three items: (i) the mathematical lemmas below (tail bounds, inclusion property); (ii) the documented outward-rounding semantics of the interval arithmetic employed (`mpmath.iv` rounds interval endpoints outward, with guard digits for the complex transcendental functions [23, 24]); and (iii) the public source code of the scripts. Item (ii) is replaceable: any outward-rounded interval implementation re-executing the same block lists obtains the same verdicts. All scripts are plain Python 3.12 + `mpmath` [23] (standard library `fractions` for the exact algebra), run independently of one another.

2.5.1 The inclusion property and tail bounds

Lemma 2.14 (Inclusion property, [25]). *Let E be any expression built from interval-enclosed constants, the variable, interval arithmetic operations, and interval elementary functions, all with outward-rounded semantics. Then for every interval box I in its domain, $E(I)$ contains the true value $E(\tau)$ for every $\tau \in I$.*

Proof. Induction on the structure of E ; see [5, Lemma 6.1]. $\square$

Lemma 2.15 (Tail bound for eta-quotient products). *Let $r := \overline{|q|} < 1$ be a certified upper bound. For $d \geq 1$, $e \in \mathbb{Z}$ and the principal branch of $\log$,*

$$\left| \log \prod_{n>N} (1 - q^{dn})^e \right| \leq E := \frac{|e| r^{d(N+1)}}{(1 - r^d)(1 - r)},$$

so the truncated product carries the rigorous complex-box tail factor $\exp(\{u + iv : |u|, |v| \leq E\})$.

Proof. For $|z| \leq r < 1$, $|\log(1 - z)| \leq |z|/(1 - r)$; sum the geometric series, cf. [5, Lemma 6.2]. Products and quotients of such factors (the parametrization (3) is a quotient of two products) are enclosed numerator- and denominator-wise before division. $\square$

On the certified paths $r \leq e^{-2\pi \cdot 0.702} < 0.0122$, and with truncation $N = 40$ the tail width satisfies $E < 10^{-78}$, far below the working precision.

2.5.2 cert2: path certification

A *path certificate* for a path $\gamma \subset \mathbb{H}$ is a finite list of parameter blocks I_ν covering γ (down to the endpoint, if the endpoint lies in V), with complex intervals Z_ν such that $c(I_\nu) \subset Z_\nu$ and every Z_ν is certified disjoint from Ast — i.e. the interval lower bound of $|\text{Re}z|^{2/3} + |\text{Im}z|^{2/3}$ over Z_ν exceeds $4^{2/3}$, a pure interval endpoint comparison. By Lemma 2.14 and Lemma 2.15 this is a proof, not an estimate (cf. [5, Prop. 6.4]).

Remark 2.16 (Generation and termination of the adaptive search). Blocks are found by adaptive bisection: accept a block when its enclosure is certified outside Ast , else bisect. Termination is not part of the trusted base, but it is guaranteed: the natural interval extension of an expression built from Lipschitz non-singular operations on a box has width $O(\text{diam})$ [25]; the factors of (3) are non-singular ($|q| < 1$, $|1 \pm q^{dn}|$ bounded away from 0) and the tail width is $< 10^{-78}$ uniformly; and on each compact path piece the true image has positive distance to Ast . Once the enclosure width drops below that margin, the block is accepted. (For a path ending at a point of $c^{-1}(K)$ — which does *not* occur in this paper — the same machinery certifies down to a cap $(y_0, y_0 + \varepsilon_0]$, closed analytically by a first-order Taylor expansion with a Cauchy-estimate remainder and a certified enclosure of the derivative at the endpoint; see [5, §4.3]. Theorems 1.2 and 1.6 have their endpoints strictly inside V_4 , so no cap is needed.)

2.5.3 cert0: exact algebraic locks

The exactness of $s_4(\tau_0) = s$ follows a four-step pattern: (i) *integrality*: $s_4 \in \mathbb{Z}((q))$ (exact integer check), so by Shimura’s CM theorem [12] the value of the eta-quotient (a modular unit with integral q -expansion) at a CM point is an algebraic integer in the corresponding ring class field; the class number bounds the degree, and conjugates are located via the level structure; (ii) *symmetric functions*: interval locks with rigorous tails force the elementary symmetric functions of the conjugate

values (e.g. sum S and product P for a quadratic value) to be explicit integers; (iii) *pinning*: an interval enclosure of the target with radius smaller than half the separation of the candidate roots singles out a unique root; (iv) *reality*: real q at the chosen points ($q(iy) > 0$, $q((1+iy)/2) < 0$) makes the products real and simplifies the locks. No exact equality is ever inferred from a floating-point approximation alone: each lock is a discreteness argument over an explicit finite candidate set.

2.5.4 verify_P1: the three-track CM evaluation

The evaluation $EK_4(\tau_0) = R$ (station S5) is certified by three independent tracks, any one of which is sufficient in principle:

- **[X]/[S] exact algebra.** The lattice decomposition of $-T_1 + 4T_2$ into Hecke and Dirichlet sums, the inclusion–exclusion coefficients, the closed forms of even-character values $L(\chi, 2)$ [16], the functional-equation constants, and the final coefficient comparison in a rational *e-basis* are all carried out in rational/algebraic-integer arithmetic (`fractions.Fraction`); no interval estimates enter. The anchors of this track are quoted standard inputs listed explicitly in each application section (Hecke’s theorem on grössencharacter theta series [13], class numbers, Dirichlet functional equations, Shimura’s CM theorem).
- **[V] interval locks.** The same identity is re-proved by outward-rounded interval arithmetic: Poisson row sums with $\coth$ / $\tanh$ closed rows and rigorous Fourier tails for the lattice sums, E_1 -continued-fraction-bracketed Mellin integrals for the L -values, Euler–Maclaurin for the Dirichlet values. Achieved half-widths are quoted in each application section (down to 9.1×10^{-53}).
- **mp cross-checks.** Independent 60-digit (`mpmath`, non-interval) reimplementations of every quantity, agreeing typically to 10^{-60} ; these corroborate the implementation but do not enter the proof.

Newform identifications use Ligozat’s eta-quotient criteria [14] (weight, character, cusp orders), exact-integer theta identity checks to order q^{60} against the Sturm bound [15] (which is ≤ 6 in all our cases), and the LMFDB labels [22]. Level and root number of each twist are determined by the *Fricke ratio* $f(i/(\sqrt{N}y)) / (y^3 f(iy/\sqrt{N}))$ evaluated at two ordinates — see §5.4 for why the naive “functional-equation residue” test is vacuous.

2.5.5 Script inventory

script	content
<code>cert0_s4_s7pair.py</code>	exactness $s_4(i\sqrt{7})$, $s_4(i\sqrt{7}/2) = 8292456 \pm 3132675\sqrt{7}$ (Theorem A)
<code>verify_P1_n4_s7pair.py</code>	Theorem A: exact separation track [S1]–[S10], FE proofs (§3.3), interval locks [V0]–[V6]
<code>cert0_s4_m144.py</code>	exactness $s_4(\tau_1) = -144$ (Theorem B)
<code>cert2_path_n4_m144.py</code>	anchor disk + certified path for Theorem B (template for Theorem F)
<code>verify_P1_n4_m144.py</code>	Theorem B: 32 checks, four tracks including [G] Sturm-level newform identification and [V] interval locks
<code>n4_m144_true_mahler.py</code>	true Mahler side of Theorem B by spectral torus integration (22-digit collision)
<code>cert0_n4_p3_t1t2.py</code>	exactness of the s_4 -values of Theorems C, D and of the conjugate of C (Theorem F's s -value)
<code>verify_P1_n4_p3_t1t2.py</code>	Theorems C, D: three tracks, [G] identifications, Fricke-ratio level determinations
<code>cert0_n4_p4_t3.py</code>	Theorem E: four-point quartic lock of the s_4 -value
<code>verify_P1_n4_p4_t3.py</code>	Theorem E: three tracks, genus-character decomposition, [V] locks
<code>n4_p4_t4_cert.py</code>	certified path for Theorem F (interior-point verdict, margin 0.11590)
<code>n4_p4_t4_verify.py</code>	Theorem F: three tracks, shifted coth / tanh row machine, [V] locks

All computations use `mpmath` interval arithmetic at 40–70 decimal digits (`iv.dps`); the certification scripts print PASS/FAIL per check and terminate with an all-checks-passed line, re-verified by the author.

3 Application I: the $\sqrt{7}$ pair

Both CM points are pure imaginary and deep inside Samart's region:

$$\tau_A = i\sqrt{7} \quad (\text{Im}\tau_A = \sqrt{7} > 1/\sqrt{2}), \quad \tau_B = \frac{i\sqrt{7}}{2} \quad (\text{Im}\tau_B = \frac{\sqrt{7}}{2} > 1/\sqrt{2}),$$

so Theorem 2.12 applies directly: station S3 is free. The work is S4 (the exact s_4 -values) and S5 (the exact CM evaluations, *separately* for the two points — an earlier stage of this project had proved only the sum of the two identities; the ray-class decomposition below closes each point independently).

3.1 Station S4: the exact s_4 -values

By `cert0_s4_s7pair.py` (four-step lock of §2.5.3): the two values are conjugate algebraic integers in $\mathbb{Q}(\sqrt{7})$ (Shimura integrality; $q(\tau_A) > 0$, $q(\tau_B) > 0$); interval locks force the symmetric functions to integers,

$$S = 16584912, \quad P = 69257922561 = 3^{12} \cdot 19^4 \quad (\text{exact check: } 8292456^2 - 7 \cdot 3132675^2 = P),$$

and the root separation 1.6×10^7 dwarfs the lock radii $|s_4(i\sqrt{7}) - r_+| \leq 9.6 \times 10^{-51}$, $|s_4(i\sqrt{7}/2) - r_-| \leq 3.6 \times 10^{-54}$. Hence exactly

$$s_4(i\sqrt{7}) = 8292456 + 3132675\sqrt{7} =: r_+, \quad s_4(i\sqrt{7}/2) = 8292456 - 3132675\sqrt{7} =: r_-.$$

3.2 Station S5: the lattice decomposition

Write $\varpi = (1 + \sqrt{-7})/2$, so $2\varpi = 1 + \sqrt{-7}$, $K = \mathbb{Q}(\sqrt{-7})$, $h(-7) = 1$, $\mathcal{O}_K^\times = \{\pm 1\}$. The lattices of (7) at the two points are (exact set equalities, checks [S1]–[S2] of `verify_P1_n4_s7pair.py`):

- at τ_A : $\Lambda_1(\tau_A) = \mathbb{Z} + i\sqrt{7}\mathbb{Z} = \mathcal{O}^{(2)}$ (the order of conductor 2, discriminant -28), and $\Lambda_2(\tau_A) = \mathcal{O}^{(4)}$ (conductor 4, discriminant -112);
- at τ_B : $\Lambda_2(\tau_B) = \mathcal{O}^{(2)}$, and $\Lambda_1(\tau_B) = \frac{1}{2}\Lambda'$, where $\Lambda' = 2\mathbb{Z} + i\sqrt{7}\mathbb{Z}$ is the index-2 *non-ideal* sublattice of $\mathcal{O}^{(2)}$; by (-4) -homogeneity, $T_1(\tau_B) = 16 T(\Lambda')$.

With $B(\Lambda) = \sum'_{\lambda \in \Lambda} |\lambda|^{-4}$, $G(\Lambda) = \sum' \bar{\lambda}^2 |\lambda|^{-6}$ (real: all three lattices are conjugation-stable), $T = 2G + B$, the ray-class decompositions

$$\begin{aligned}\mathcal{O}^{(2)} \setminus \{0\} &= \{\alpha \equiv 1 \pmod{2}\} \sqcup 2\mathcal{O}_K \setminus \{0\}, \\ \mathcal{O}^{(4)} &= 2\mathcal{O}^{(2)} \sqcup \{\alpha \equiv 1, 3 \pmod{4}\}, \\ \Lambda' &= 2\mathcal{O}^{(2)} \sqcup \{\alpha \equiv 1+2\varpi, 3+2\varpi \pmod{4}\},\end{aligned}$$

the inclusion–exclusion coefficients $9/16$ and $23/16$ at the split prime $(2) = (\varpi)(\bar{\varpi})$ (exact, using $\varpi^2 + \bar{\varpi}^2 = -3$), and the mod-4 ray-class pairing of the character $\chi_{-4} \circ N$ (values $+1, +1, -1, -1$ on the four unit classes, check [X4]) give, all in exact rational arithmetic (checks [S3]–[S6]):

lattice	B	G	$T = 2G + B$
$\mathcal{O}^{(2)}$	$\frac{5}{4}\zeta_K(2)$	$3L_3$	$6L_3 + \frac{5}{4}\zeta_K(2)$
$\mathcal{O}^{(4)}$	$\frac{41}{64}\zeta_K(2) + z_V$	$\frac{13}{8}L_3 + L_3^{\text{tw}}$	$\frac{13}{4}L_3 + 2L_3^{\text{tw}} + \frac{41}{64}\zeta_K(2) + z_V$
Λ'	$\frac{41}{64}\zeta_K(2) - z_V$	$\frac{13}{8}L_3 - L_3^{\text{tw}}$	$\frac{13}{4}L_3 - 2L_3^{\text{tw}} + \frac{41}{64}\zeta_K(2) - z_V$

where $L_3 := L(g_7, 3)$, $L_3^{\text{tw}} := L(g_7 \otimes \chi_{-4}, 3)$, and $z_V := L(\chi_{-4}, 2)L(\chi_{-28}, 2)$. The pointwise assemblies (check [S8], including the separation witness `combA` $\neq$ `combB`):

$$\begin{aligned}\text{combA} &:= -T(\mathcal{O}^{(2)}) + 4T(\mathcal{O}^{(4)}) = 7L_3 + 8L_3^{\text{tw}} + \frac{21}{16}\zeta_K(2) + 4z_V, \\ \text{combB} &:= -16 T(\Lambda') + 4T(\mathcal{O}^{(2)}) = -28L_3 + 32L_3^{\text{tw}} - \frac{21}{4}\zeta_K(2) + 16z_V.\end{aligned}$$

3.3 The functional equations and root numbers

Two functional-equation inputs are proved in `verify_P1_n4_s7pair.py`, not quoted: $M_7 = \frac{7\sqrt{7}}{4\pi^3} L_3$ [5, Lemma 2.8] and the twisted identity

$$U_{a/4} W_{112} = 4 \gamma_a W_7 U_{a/4} \quad (a = 1, 3), \quad \gamma_a = \begin{pmatrix} (1 + 7a^2)/4 & a \\ 7a & 4 \end{pmatrix} \in \Gamma_0(7), \quad (10)$$

an exact matrix identity (check [X10], with $\chi_{-7}(4) = +1$), which gives $h|[W_{112}]_3 = i h$ for $h = g_7 \otimes \chi_{-4}$ — the Atkin–Li twist- W technique of [17] made self-contained for these parameters — hence root number $+1$ at level 112 and $M_7^{\text{tw}} = \frac{112\sqrt{7}}{\pi^3} L_3^{\text{tw}}$. The remaining quoted constants are the Dirichlet functional equations (textbook) and the finite-sum evaluation

$$L(\chi_{-28}, 2) = \frac{2\pi^2}{7\sqrt{7}},$$

whose character sum $\sum_a \chi_{-28}(a)a^2 = 448$ is an exact integer check [X7] [16].

3.4 Assembly

With the e -basis $e = \frac{\sqrt{7}}{\pi^3}(L_3, L_3^{\text{tw}}, \zeta_K(2), z_V)$ one has $M_7 = \frac{7}{4}e_1$, $M_7^{\text{tw}} = 112e_2$, $d_7 = \frac{21}{2}e_3$, $d_4 = 7e_4$ (exact, [S9]); the coefficient comparison

$$\text{EK}_4(i\sqrt{7}) = 10 \text{ combA} \cdot e = \left(70, 80, \frac{105}{8}, 40\right) \cdot e = \frac{5}{28}(4M_7^{\text{tw}} + 224M_7 + 32d_4 + 7d_7),$$

$$\text{EK}_4\left(\frac{i\sqrt{7}}{2}\right) = 5 \text{ combB} \cdot e = \left(-140, 160, -\frac{105}{4}, 80\right) \cdot e = \frac{5}{14}(4M_7^{\text{tw}} - 224M_7 + 32d_4 - 7d_7),$$

is exact rational arithmetic (check [S10]), and the two right-hand sides are distinct — the separation is complete and each identity holds individually. Combined with Theorem 2.12 and §2.5.3 this proves Theorem 1.1. $\square$

Remark 3.1 (Independent confirmations). The mp track (60 dps, independent implementations: Mellin integrals, Poisson row sums, ray-class sums) confirms every step to 9×10^{-60} ; the interval-lock track [V0]–[V6] (`iv.dps=70`: continued-fraction bracketing, Fourier tail bounds, Euler–Maclaurin) locks the two final identities with half-widths 1.15×10^{-52} and 2.30×10^{-52} . Reference values:

$$\begin{aligned} M_7 &= 0.1026716077789020112104565948982929139989\dots, \\ M_7^{\text{tw}} &= 9.887687024790914246588215159482724883665\dots \end{aligned}$$

4 Application II: $n_4(-144)$

The CM point

$$\tau_1 = \frac{1 + \sqrt{-3}}{2}$$

is *not* pure imaginary, so Theorem 2.12 does not apply directly; this is the first use of the full continuation machine in the n_4 family. It also shows the machine in its shortest form: the target is a strict interior point of V_4 , so no endpoint cap and no boundary-continuity step are needed.

4.1 Station S3: the certified path

The path γ from the anchor disk $D = \{|\text{Re}\tau| \leq 1/32, |\text{Im}\tau - 1| \leq 1/32\}$ to τ_1 — horizontal at $\text{Im}\tau = 1$ to $1/2+i$, then vertical at $\text{Re}\tau = 1/2$ down to τ_1 — is certified inside V_4 by `cert2.path.n4.m144.py` (machinery of §2.5.2): the anchor disk passes as a single block with astroid margin 0.291 and the certified bound $|s_4| \geq 493 > 256$ (which activates Lemma 2.13: $F = G$ on D); the horizontal leg takes 2 blocks (minimum margin 0.148); the vertical leg takes 1 block (margin 1.051); and τ_1 itself satisfies $c(\tau_1) = 2\sqrt{3}e^{-i\pi/4}$, whose astroid functional is $3.634 > 4^{2/3} = 2.520$ — a strict exterior point of Ast , so $\tau_1 \in V_4$ itself and $\tau_1 \in W$. The endpoint Taylor cap of [5] is absent. By Theorem 2.10, $n_4(s_4(\tau_1)) = \text{EK}_4(\tau_1)$.

4.2 Station S4: the exact s_4 -value

`cert0.s4.m144.py`: $q(\tau_1) = -e^{-\pi\sqrt{3}}$ is a negative real, all products are real, and the value is pinned to a *rational integer* (Shimura integrality plus $h(-3) = 1$, so the ring class field is K ; reality from the integral q -expansion and T -invariance): $|s_4(\tau_1) + 144| \leq 2.3 \times 10^{-51} < 1/2$, hence $s_4(\tau_1) = -144$ exactly.

4.3 Station S5: the CM evaluation

$K = \mathbb{Q}(\sqrt{-3})$, $\mathcal{O}_K = \mathbb{Z}[\omega]$, $\omega = (-1 + \sqrt{-3})/2$, units μ_6 . The structural simplification is the μ_6 annihilation $\sum_{u \in \mu_6} u^2 = 0$ (exact check [X1]): it forces $G(\mathcal{O}_K) = 0$, which is also the reason $\mathbb{Q}(\sqrt{-3})$ has no conductor-(1) grössencharacter of type $(2, 0)$ and the evaluation involves a non-trivial conductor. The lattices are $\Lambda_1(\tau_1) = \mathcal{O}_K$ (since $\tau_1 = 1 + \omega = -\omega^2$) and $\Lambda_2(\tau_1) = \mathcal{O}_2$ (conductor 2; 2 is inert), giving (exact [X2]–[X8]):

$$T_1(\tau_1) = 6\zeta_K(2), \quad T_2(\tau_1) = 4L(g_{12}, 3) + \frac{9}{4}\zeta_K(2), \quad -T_1 + 4T_2 = 16L(g_{12}, 3) + 3\zeta_K(2);$$

note the $\zeta_K(2)$ -term does *not* cancel (coefficient 3) — this is the source of the d_3 -term. The theta identity $2g_{12} = \sum'_{\alpha \equiv 1(2)} \alpha^2 q^{N\alpha}$ is checked with exact integer arithmetic to order q^{60} (check [L1]); the newform identification $g_{12} = \eta(2\tau)^3 \eta(6\tau)^3 \in S_3(\Gamma_0(12), \chi_{-3})$ is certified at Sturm level (checks [G1]–[G3]): Ligozat’s criteria [14] (both congruences $24 \equiv 0 \pmod{24}$; character $-2^3 3^3$, square kernel -3), all six cusp orders equal 1 (cuspidal; divisor degree $6 = \frac{3}{12} \cdot 24$), Sturm bound $\lfloor 3 \cdot 24/12 \rfloor = 6 < 60$ [15]; LMFDB label 12.3.c.a [22]. With $M_{12} = \frac{6\sqrt{3}}{\pi^3} L(g_{12}, 3)$ (root number $+1$, Fricke ratio) and $d_3 = \frac{3\sqrt{3}}{4\pi} L(\chi_{-3}, 2)$, the e -basis assembly gives (exact, [S1]):

$$\text{EK}_4(\tau_1) = \frac{5\sqrt{3}}{\pi^3} (16L(g_{12}, 3) + 3\zeta(2)L(\chi_{-3}, 2)) = \frac{10}{3} (4M_{12} + d_3),$$

both sides being $(80, 5/2)$ in the basis $(\sqrt{3}L(g_{12}, 3)/\pi^3, \sqrt{3}L(\chi_{-3}, 2)/\pi)$. Theorem 1.2 follows. $\square$

Remark 4.1 (Confirmations and the true-Mahler end). The mp track confirms the identity to 3.7×10^{-60} (check [E1]); the interval-lock track [V0]–[V3] (shifted coth / tanh rows, since $\text{Re}\tau_1 = 1/2$) locks it with half-width 5.8×10^{-52} . Independently, direct spectral torus integration (`n4_m144_true_mahler.py`: $c(\tau_1)$ lies outside Ast, so the Jensen integrand is real-analytic periodic and the two-dimensional trapezoidal rule converges spectrally; residual 1.16×10^{-22} at $N = 128$) collides the two sides of Theorem 1.2 to 22 digits:

$$n_4(-144) = 5.09841965623737833323 \dots$$

The propagation argument does not depend on this numerical value. Reference value: $M_{12} = 0.3016149874129407464690529311477683998854 \dots$

5 Application III: class number one, deep interior points (Theorems C and D)

Both CM points are pure imaginary and deep inside Samart’s region,

$$\tau_C = i\sqrt{2} \quad (\text{Im}\tau_C = \sqrt{2}), \quad \tau_D = 2i \quad (\text{Im}\tau_D = 2),$$

so Theorem 2.12 applies directly and the proofs consist of `cert0` locks and three-track CM evaluations. The new methodological content of this section is concentrated in station S5: the character $\chi_8 \circ N$ as an exact mod-2 ray-class projector, and the level determination of the twists, which contains a genuine trap.

5.1 The ray-class projector $\chi_8 \circ N$

In both fields the twist character materializes as a parity condition. Exact checks ([X4], [Y4] of `verify_P1_n4_p3_t1t2.py`):

- in $K = \mathbb{Q}(\sqrt{-2})$: for $a + b\sqrt{-2}$ of odd norm, $\chi_8(N(a + b\sqrt{-2})) = +1 \iff b \text{ even } (a \text{ odd})$;
- in $K = \mathbb{Q}(i)$: for $a + bi$ of odd norm, $\chi_8(N(a + bi)) = +1 \iff b \equiv 0 \pmod{4} \text{ } (a \text{ odd, } b \text{ even})$.

Hence $(1 + \chi_8 \circ N)/2$ is exactly the projector onto the mod-2 ray class $\{\alpha \equiv 1 \pmod{2}\}$, and in class number one no genus character is needed: every ideal decomposition in this section is a parity bookkeeping exercise certified in rational arithmetic. Also used everywhere below: the closed form

$$L(\chi_8, 2) = \frac{\pi^2 \sqrt{2}}{16}$$

(even-character finite-sum formula [16]: $\tau(\chi_8) = 2\sqrt{2}$, $\sum_a \chi_8(a)a^2 = 16$, exact check [X7]).

5.2 Theorem C: $\tau_C = i\sqrt{2}$

Station S4. `cert0_n4_p3_t1t2.py`: $q(i\sqrt{2}) = e^{-2\pi\sqrt{2}} > 0$ and $q((1 + i\sqrt{2})/2) = -e^{-\pi\sqrt{2}} < 0$ are real, so real interval locks suffice; the symmetric functions lock to the integers

$$S = 7312, \quad P = -153664 = 3656^2 - 2 \cdot 2600^2$$

($|\Delta| \leq 1.1 \times 10^{-49}$, 5.2×10^{-48} ; root separation 7354), pinning

$$s_4(i\sqrt{2}) = 3656 + 2600\sqrt{2}, \quad s_4\left(\frac{1 + i\sqrt{2}}{2}\right) = 3656 - 2600\sqrt{2}$$

— the second value is reused in Theorem 1.6.

Station S5. $K = \mathbb{Q}(\sqrt{-2})$, $\mathcal{O}_K = \mathbb{Z}[\sqrt{-2}]$, $h = 1$, units ± 1 , and $2 = -(\sqrt{-2})^2$ ramifies. The lattices are $\Lambda_1 = \mathcal{O}_K$ and $\Lambda_2 = \mathcal{O}_2 = \mathbb{Z} + 2\mathcal{O}_K$. With $L_8 := L(g_8, 3)$, $L_8^{\text{tw}} := L(g_8 \otimes \chi_8, 3)$, and $z_V := L(\chi_8, 2)L(\chi_{-4}, 2)$, the inclusion–exclusion at the ramified prime (factors $1/4$ for B , $-1/4$ for G) gives (exact, [S]-track):

$$T(\mathcal{O}_K) = 4L_8 + 2\zeta_K(2), \quad T(\mathcal{O}_2) = \frac{11}{4}L_8 + 2L_8^{\text{tw}} + \frac{7}{8}\zeta_K(2) + z_V,$$

$$\text{comb} := -T_1 + 4T_2 = 7L_8 + 8L_8^{\text{tw}} + \frac{3}{2}\zeta_K(2) + 4z_V.$$

The newform identification $g_8 = \eta(\tau)^2\eta(2\tau)\eta(4\tau)\eta(8\tau)^2 \in S_3(\Gamma_0(8), \chi_{-8})$ is Sturm-level ([G]-track: Ligozat congruences $24 \equiv 0 \pmod{24}$; cusp orders $(1, \frac{1}{2}, \frac{1}{2}, 1)$; weighted divisor degree $3 = \frac{3}{12} \cdot 12$; Sturm bound 3; theta identity $2g_8 = \sum'_{\alpha \in \mathcal{O}_K} \alpha^2 q^{N\alpha}$ exact to q^{60}). In the e -basis $e = \frac{\sqrt{2}}{\pi^3}(L_8, L_8^{\text{tw}}, \zeta_K(2), z_V)$:

$$M_8 = 4e_1, \quad M_8^{\text{tw}} = 32e_2 \quad [\text{level } 32, w = +1, \S 5.4], \quad d_8 = 24e_3, \quad d_4 = 16e_4,$$

and the exact comparison ([S3]) $\text{EK}_4(i\sqrt{2}) = 10 \text{ comb} \cdot e = (70, 80, 15, 40) \cdot e$ is the right-hand side of Theorem 1.3. $\square$

5.3 Theorem D: $\tau_D = 2i$

Station S4. The same script locks

$$S = 286416, \quad P = -126023688 = 143208^2 - 2 \cdot 101574^2$$

($|\Delta| \leq 4.1 \times 10^{-48}$, 3.7×10^{-45} ; separation 2.87×10^5), pinning $s_4(2i) = 143208 + 101574\sqrt{2}$ and $s_4((1+2i)/2) = 143208 - 101574\sqrt{2}$ (the latter is the open entry #1 of Table 1).

Station S5. $K = \mathbb{Q}(i)$, $\mathcal{O}_K = \mathbb{Z}[i]$, $h = 1$, units μ_4 — and $\sum_{u \in \mu_4} u^2 = 0$, so $G(\mathcal{O}_K) = 0$ (exact [Y1]; same annihilation mechanism as Theorem 1.2); $2 = -i(1+i)^2$ ramifies. The lattices are $\Lambda_1 = \mathcal{O}_2$ and $\Lambda_2 = \mathcal{O}_4$. Every ideal coprime to 2 has exactly two generators $\equiv 1 \pmod{2}$ ([Y3]), so the mod-2 ray sums carry *no twist term*: with $L_{16} := L(g_{16}, 3)$, $L_{16}^{\text{tw}} := L(g_{16} \otimes \chi_8, 3)$, and $z_{V2} := L(\chi_8, 2)L(\chi_{-8}, 2)$,

$$T(\mathcal{O}_2) = 4L_{16} + \frac{7}{4}\zeta_K(2), \quad T(\mathcal{O}_4) = \frac{9}{4}L_{16} + 2L_{16}^{\text{tw}} + \frac{55}{64}\zeta_K(2) + z_{V2},$$

$$\text{comb} = 5L_{16} + 8L_{16}^{\text{tw}} + \frac{27}{16}\zeta_K(2) + 4z_{V2},$$

where the $\mathcal{O}_4$ decomposition uses $\mathcal{O}_4 \setminus \{0\} = \{a \text{ odd}, b \equiv 0 \pmod{4}\} \sqcup 2\mathcal{O}_2 \setminus \{0\}$ and the projector of §5.1 ([Y4]). The newform identification $g_{16} = \eta(4\tau)^6 \in S_3(\Gamma_0(16), \chi_{-4})$ is Sturm-level (five cusp orders all 1; weighted degree $6 = \frac{3}{12} \cdot 24$; Sturm bound 6; theta identity $2g_{16} = \sum_{\alpha \equiv 1 \pmod{2}} \alpha^2 q^{N\alpha}$ of conductor (2) exact to q^{60}). In the e -basis $e = \frac{1}{\pi^3}(L_{16}, L_{16}^{\text{tw}}, \zeta_K(2), z_{V2})$:

$$M_{16} = 16e_1, \quad M_{16}^{\text{tw}} = 128e_2 \quad [\text{level } 64, w = +1], \quad d_4 = 12e_3, \quad d_8 = 64e_4,$$

and the exact comparison ([S6]) $\text{EK}_4(2i) = 20 \text{ comb} \cdot e = (100, 160, 135/4, 80) \cdot e$ is the right-hand side of Theorem 1.4. $\square$

5.4 The twist-level trap and the Fricke ratio

Two methodological points emerged here and now belong to the machine.

(i) The “FE residue” test is vacuous. A widespread heuristic decides the root number of a completed L -function $\Lambda(f, s) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(f, s)$ by splitting the Mellin integral at $y = 1$ and comparing the two sides of $\Lambda(1) \stackrel{?}{=} w\Lambda(2)$. But the split gives $\Lambda(s) = I_1(s) + wI_2(3-s)$ -type expressions for which, at $w = +1$, the two sides are *the same linear combination of the same two integrals at every candidate level N* : the “residue” vanishes identically in N and certifies nothing. (The same vacuity affected a check in the $\sqrt{7}$ -pair script; there the conclusion was independently secured by the Fricke ratio, so no result was endangered.)

(ii) The non-vacuous discriminator is the Fricke ratio. For the true level N_0 one has the exact transformation $f(i/(\sqrt{N_0}y)) = w y^3 f(iy/\sqrt{N_0})$ for *all* $y > 0$, coming from the Fricke involution (proved for each form in this paper either by the eta transformation law or by a twisted Fricke matrix identity such as (10)). Hence the ratio

$$R_N(y) := \frac{f(i/(\sqrt{N}y))}{y^3 f(iy/\sqrt{N})}$$

is identically w at $N = N_0$, while at $N \neq N_0$ it varies with y ; evaluation at two ordinates separates the candidates. This is how the levels in this paper were fixed: 32 for $g_8 \otimes \chi_8$, 64 for $g_{16} \otimes \chi_8$ (two-ordinate agreement to 3.1×10^{-61} and better), and 96 for the twists of Section 6 (unique match in the candidate scan $\{24, 48, 96, 192, 384\}$).

(iii) **The trap itself.** For $g_{16} \otimes \chi_8$ the naive level guess $\text{lcm}(16, 4^2)$ -style reasoning suggests 32 — and the vacuous test (i) happily “confirms” it. With the wrong level-32 Mellin constant, M_{16}^{tw} comes out smaller by a factor $2\sqrt{2}$, and the conjectured identity fails numerically by $\sim 1\%$; a first pass over Theorem 1.4 recorded exactly this mismatch, briefly suggesting an $s = 81$ -type refutation. The Fricke ratio at two ordinates gives the true level 64, and with it the identity holds to 2.5×10^{-60} . Since a wrong level *silently scales* an L -value by a simple algebraic factor, we record the rule: *never determine a twist level by a residual; determine it by the Fricke ratio at two ordinates, and cross-check against the assembled identity.*

Remark 5.1 (Confirmations for Theorems C and D). The mp tracks (60 dps) confirm the two final identities to 7.5×10^{-60} and 5×10^{-60} ; the interval-lock tracks [V] contain zero with half-widths $\leq 4.6 \times 10^{-53}$ (Theorem C) and $\leq 6.2 \times 10^{-53}$ (Theorem D). Direct torus integration ($c \approx 9.25$ and 23.14, smooth integrands) confirms both at float64 level. Reference values:

$$\begin{aligned} M_8 &= 0.1353184954231269106150060654595395883532\dots, \\ M_8^{\text{tw}} &= 1.549084847095611486271658159444376281785\dots \quad (\text{level } 32, w = +1), \\ M_{16} &= 0.4975478545679851929276297713583412482363\dots, \\ M_{16}^{\text{tw}} &= 4.336361249244445609135012019721278212973\dots \quad (\text{level } 64, w = +1), \\ d_8 &= \frac{4\sqrt{2}}{\pi} L(\chi_{-8}, 2) = 1.9171950931209540617988237536697845644585\dots \end{aligned}$$

(d_8 by the functional equation and by direct numerical differentiation, agreeing exactly at working precision).

6 Application IV: class number two (Theorem E)

The CM point $\tau_E = i\sqrt{6}$ is pure imaginary with $\text{Im}\tau_E = \sqrt{6} > 1/\sqrt{2}$, so Theorem 2.12 applies directly. The field $K = \mathbb{Q}(\sqrt{-6})$ has discriminant -24 and *class number 2* — the first such case treated by the machine — and everything is one step richer: two newforms, a genus character, a non-principal ideal class, and a four-point `cert0` lock.

6.1 Station S4: the four-point quartic lock

Class number two makes the s_4 -value quartic over $\mathbb{Q}$, in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. `cert0_n4.p4.t3.py` locates and locks all four conjugate values:

signs	CM point	lock half-width
(+, +)	$i\sqrt{6} \ (q = e^{-2\pi\sqrt{6}} > 0)$	7.0×10^{-47}
(-, +)	$i\sqrt{6}/4 \ (q = e^{-\pi\sqrt{6}/2} > 0)$	4.4×10^{-51}
(+, -)	$(1 + i\sqrt{6})/2 \ (q = -e^{-\pi\sqrt{6}} < 0)$	3.1×10^{-50}
(-, -)	$1/2 + i\sqrt{6}/4 \ (q = -e^{-\pi\sqrt{6}/2} < 0)$	6.8×10^{-53}

with minimal polynomial

$$X^4 - 4829472X^3 - 8676927360X^2 + 3042408646656X + 7520406736896,$$

whose coefficients are confirmed as integers both by exact $\mathbb{Z}[\sqrt{2}, \sqrt{3}]$ arithmetic and by the interval locks; the minimum root separation 305.1 dwarfs every lock radius. Two structural observations aided the location (used for location only, not in the certified chain): s_4 is W_2 -invariant,

$s_4(-1/(2\tau)) = s_4(\tau)$, observed numerically to 10^{-33} at sample points; and the second class of discriminant -24 degenerates: $s_4(i\sqrt{6}/2) = s_4(i\sqrt{6}/6) = 2304 \in \mathbb{Z}$, which is why that value is absent from the quartic roots. In particular the $(+, -)$ value is open entry #2 of Table 1 and the $(-, \pm)$ values are entries #3, #4: their station S4 is already done.

6.2 The two newforms and the genus character

The class group is $\mathbb{Z}/2$ and coincides with the genus group. The theta series of the two grössencharacters ψ and $\psi \cdot \chi_g$ of type $(2, 0)$ have coefficients (exact-integer checks [th0]–[th6]: integrality, $a(1) = 1$, the Hecke recursion $a(p^2) = a(p)^2 - \chi_{-24}(p)p^2$, multiplicativity, vanishing at inert primes, distinctness):

$$a^{(1)}(n) = P(n) - Q(n), \quad a^{(2)}(n) = P(n) + Q(n),$$

$$P(n) = \frac{1}{2} \sum_{a^2+6b^2=n} ' (a^2 - 6b^2), \quad Q(n) = \frac{1}{4} \sum_{2x^2+3y^2=n} ' (4x^2 - 6y^2),$$

P collecting the principal ideal class and Q the non-principal one (elements of norm $2n$ in $\mathfrak{p}_2 = (2, \sqrt{-6})$ are in bijection with non-principal ideals of norm n). By Hecke's theorem [13] both are newforms of $S_3(\Gamma_0(24), \chi_{-24})$, which has dimension 2, so they exhaust the space; the coefficient match $(q \mp 2q^2 \pm 3q^3 + 4q^4 \pm 2q^5 - 6q^6)$ fixes the LMFDB labels

$$g_{24}^{(1)} = P - Q = 24.3.\text{h.a.}, \quad g_{24}^{(2)} = P + Q = 24.3.\text{h.b.},$$

the unique rational CM newforms of level 24 and weight 3 [22] (in agreement with Schütt's classification [20]). Samart's $M_{24}^{(1)}$ (coefficient 28 in Theorem 1.5) is our $g_{24}^{(1)} = P - Q$; swapping the labels breaks the identity at the 0.4 level, a labeling pitfall we record. The *genus character* is $\chi_g = \chi_8 \circ N = \chi_{-3} \circ N$ on ideal norms (exact check [th2]: $\chi_8 \chi_{-3} = \chi_{-24}$, the field character, so the two compositions coincide on norms), and it handles exactly the non-principal class that the mod-2 ray projector of §5.1 cannot see.

6.3 Station S5: the class-number-two lattice decomposition

The lattices are $\Lambda_1 = \mathcal{O}_K$ and $\Lambda_2 = \mathcal{O}_2$. The new ingredients, relative to class number one, are: (a) the class-group projector $(1 + \chi_g)/2$ on the B -side and the difference of the two ψ -values on the G -side; (b) sums over $\mathfrak{p}_2$ as a lattice, with homothety factors $1/4$ (not $1/2$ — numerically locked, then certified); (c) the ray condition $\{a \text{ odd}, b \text{ even}\}$ via $\chi_{-8} \circ N$, which is *not* a Hecke character (nontrivial on principal ideals) but whose L_K -value still factors as a product of Dirichlet L -functions. Exact decomposition:

$$B(\mathcal{O}_K) = \zeta_K(2) + L(\chi_8, 2)L(\chi_{-3}, 2), \quad G(\mathcal{O}_K) = L(g_{24}^{(1)}, 3) + L(g_{24}^{(2)}, 3),$$

$$B(\mathfrak{p}_2) = \frac{\zeta_K(2) - L(\chi_8, 2)L(\chi_{-3}, 2)}{4}, \quad G(\mathfrak{p}_2) = \frac{L(g_{24}^{(2)}, 3) - L(g_{24}^{(1)}, 3)}{4},$$

(the sign in $G(\mathfrak{p}_2)$ is $\psi(\mathfrak{p}_2) = -2$), and

$$B_{\text{odd}, \chi} = L(\chi_{-8}, 2)L(\chi_3, 2) + L(\chi_{-4}, 2)L(\chi_{24}, 2), \quad G_{\text{odd}, \chi} = L(g_{24}^{(1)} \otimes \chi_{-8}, 3) + L(g_{24}^{(2)} \otimes \chi_{-8}, 3).$$

Note $T(\Lambda) = B(\Lambda) + 2G(\Lambda)$ for *every* lattice: the pointwise identity $4(\text{Re } z)^2/|z|^6 - 1/|z|^4 = 2\text{Re}(z^2)/|z|^6 + 1/|z|^4$ (exact check [X0]) plus absolute convergence, closing what had briefly appeared to be a gap. The even primitive closed forms [16]

$$L(\chi_3, 2) = \frac{\pi^2 \sqrt{3}}{18} \quad (\text{conductor } 12, \sum a^2 \chi_3(a) = 48), \quad L(\chi_{24}, 2) = \frac{\pi^2 \sqrt{6}}{24} \quad (\sum a^2 \chi_{24}(a) = 288)$$

are absorbed by the d_8 and d_4 terms in the assembly. With $L_1, L_2, L_1^{\text{tw}}, L_2^{\text{tw}}$ the four L -values at $s = 3$,

$$\begin{aligned} \text{comb} &= \frac{7}{2}L_1 + \frac{3}{2}L_2 + 4L_1^{\text{tw}} + 4L_2^{\text{tw}} + \frac{3}{4}\zeta_K(2) \\ &\quad + \frac{7}{4}L(\chi_8, 2)L(\chi_{-3}, 2) + 2L(\chi_{-8}, 2)L(\chi_3, 2) + 2\text{Cat } L(\chi_{24}, 2), \end{aligned}$$

and the [S1]–[S8] Fraction-exact assembly (single-term substitutions) produces the coefficient list

$$\frac{35}{12}, \frac{5}{4}, \frac{5}{12}, \frac{5}{12}, \frac{35}{12}, \frac{5}{2}, \frac{5}{6}, \frac{5}{48},$$

yielding Theorem 1.5. □

Remark 6.1 (Traps recorded for the machine). (i) χ_3 has conductor 12 with period values 0 at even positions; a wrongly tabulated period (nonzero even positions) makes standard Dirichlet- L software silently return a wrong value (1.1325 instead of 0.951...), which once produced a false failure at the 3.5×10^{-2} level. The machine now treats character tables as exact data to be certified ([X3]). (ii) The twist levels (96) were fixed by the Fricke-ratio scan of §5.4, never by residues.

Remark 6.2 (Confirmations). The mp track (60 dps) confirms the final identity to 2.49×10^{-60} (check [E]); the [V] interval-lock track (Dual-number coth rows, E_1 -bracketed Mellin with the self-contained coefficient bound $|a(n)| \leq 6n^3$, level-96 Mellin truncation $n_0 \approx 481$) locks $T(\mathcal{O}_K)$, $T(\mathcal{O}_2)$, and the final identity with half-widths 4.50×10^{-53} , 4.92×10^{-53} , and 9.72×10^{-53} . Direct torus integration ($c \approx 46.88$, smooth) confirms at float64 level. Reference values (50 dps, Mellin; level 96, $w = +1$ for the twists):

$$\begin{aligned} M_{24}^{(1)} &= 0.8366946514850983661417153108277990376289\dots, \\ M_{24}^{(2)} &= 1.093580625535742995623816182417982487423\dots, \\ M_{24}^{(1),\text{tw}} &= 8.547412838595950972260990976022241348897\dots, \\ M_{24}^{(2),\text{tw}} &= 7.183569611977336512537940485577907159979\dots \end{aligned}$$

7 Application V: the boundary point that turned out interior (Theorem F)

The point

$$\tau_4 = \frac{1 + \sqrt{-2}}{2}, \quad \text{Im}\tau_4 = \frac{1}{\sqrt{2}},$$

lies *exactly* on the boundary of Samart’s applicability region. Moreover $q(\tau_4) = -e^{-\pi\sqrt{2}} < 0$ is a negative real (outside his $q \in (0, 1)$ hypothesis) and $|s_4(\tau_4)| = 20.96\dots < 256$ (outside the ${}_5F_4$ convergence region), so Theorem 2.12 fails to cover τ_4 for two independent reasons. This was expected to require a boundary-continuity argument. It does not: the boundary in question is not the boundary of the problem.

7.1 Samart’s boundary is not the topological boundary of V_4

A 60-digit scan along the line $\text{Re}\tau = 1/2$ shows the astroid functional $|\text{Rec}|^{2/3} + |\text{Imc}|^{2/3}$ of $c(\tau)$ decreasing monotonically from 4.377 at $y = 1$, crossing the threshold $4^{2/3} = 2.51984$ only at

$y \approx 0.693$ (margin $+0.064$ at $y = 0.700$, -0.011 at $y = 0.690$). At τ_4 ($y = 1/\sqrt{2} = 0.70711$) the functional is 2.6357 , with margin $+0.116$. Since c is holomorphic (an open map) and Ast is closed, τ_4 is a *strict interior point* of V_4 . Samart's line $\{\text{Im}\tau = 1/\sqrt{2}\}$ is dictated by the real- q mechanism of his proof; the topological boundary of V_4 along this segment lies ≈ 0.014 lower, and τ_4 sits strictly between the two. Consequently the short `cert2` path of Theorem 1.2 suffices, and no boundary-continuity step is needed.

7.2 Station S3: the certified path

`n4_p4_t4_cert.py` (template `cert2_path_n4_m144.py`; leg B extended down to $y = 0.702 < 1/\sqrt{2}$ so that τ_4 lies strictly inside the certified leg; the $|q|$ bound 0.0122 keeps the tail $r^{41} < 10^{-78}$):

- anchor disk D : margin 0.29143 , certified $|c| \geq 4.71$, $|s_4| \geq 493.6 > 256$ (activating Lemma 2.13);
- leg A ($y = 1$, $x \in [0, 1/2]$): 2 blocks, minimum margin 0.14759 ;
- leg B ($x = 1/2$, $y \in [0.702, 1]$): 4 blocks, minimum margin 0.02270 (most dangerous block $y \in [0.702, 0.7393]$);
- point enclosure at τ_4 : astroid margin $0.11590 > 0$ — an interval-certified interior-point verdict;
- self-tests: 21/21 point inclusions (80-dps values inside interval enclosures, including τ_4 itself) and 5/5 monotonicity checks.

Hence the whole path and τ_4 lie in W , the component of the anchor disk, and Theorem 2.10 gives $n_4(s_4(\tau_4)) = \text{EK}_4(\tau_4)$. As a byproduct, the open entries #1 ($\tau = (1 + 2i)/2$, the top corner of the legs) and #7 ($\tau = (3 + i\sqrt{21})/6 = 1/2 + 0.7638i$, on leg B) of Table 1 already have their station S3 certified by this same path.

7.3 Station S4

The value $s_4(\tau_4) = 3656 - 2600\sqrt{2}$ is already locked exactly by `cert0_n4.p3_t1t2.py` as the conjugate point of Theorem 1.3 ($S = 7312$, $P = -153664$; §5.2).

7.4 Station S5: a different lattice decomposition, not a change of clothes

Although the s -value is conjugate to Theorem 1.3's, the CM point is different, and the lattice structure differs: $2\tau_4 = 1 + \sqrt{-2}$ gives $\Lambda_2 = \mathcal{O}_K$ (not $\mathcal{O}_2$), and

$$2\Lambda_1 = 2\mathbb{Z} + \mathbb{Z}(1 + \sqrt{-2}) = \{a + b\sqrt{-2} : a \equiv b \pmod{2}\} =: \mathcal{O}' = 2\mathcal{O}_K \sqcup ((1 + \sqrt{-2}) + 2\mathcal{O}_K),$$

a non-ideal sublattice (exact [X1]–[X3]). The parity-class sums from Theorem 1.3's anchors ([X4], [X5], homothety $1/16$ [X8]):

$$B_{00} = \frac{\zeta_K(2)}{8}, \quad B_{11} = \frac{3}{4}\zeta_K(2) - z_V, \quad G_{00} = \frac{L_8}{8}, \quad G_{11} = \frac{5}{4}L_8 - L_8^{\text{tw}},$$

$$T(\mathcal{O}') = \frac{11}{4}L_8 - 2L_8^{\text{tw}} + \frac{7}{8}\zeta_K(2) - z_V, \quad T(\Lambda_1) = 16T(\mathcal{O}'),$$

$$\text{comb} = -T(\Lambda_1) + 4T(\Lambda_2) = -28L_8 + 32L_8^{\text{tw}} - 6\zeta_K(2) + 16z_V.$$

Note the sign pattern: the twisted and $\zeta_K(2)$ terms flip sign relative to Theorem 1.3 (combinations $-28/+32$ versus $+7/+8$), and the $-d_8$ term traces to the coefficient -16 of z_V (exact witness [S7]: the two sign patterns are genuinely different). The e -basis assembly of §5.2 gives (exact [X10]/[S6])

$$\text{EK}_4(\tau_4) = 5 \text{ comb} \cdot e = (-140, 160, -30, 80) \cdot e = \frac{5}{4}(4M_8^{\text{tw}} - 28M_8 + 4d_4 - d_8),$$

proving Theorem 1.6. □

Remark 7.1 (Confirmations). The mp track (60 dps; the L -values recomputed in-script at levels $8/32$, cross-checked against Section 5 to 1.6×10^{-40}) confirms the final identity to 9.33×10^{-61} (check [E1]); the [V] interval-lock track — extended by a shifted coth / tanh row machine (`lattice_T_iv_t4`: at $x_0 = 1/2$ the Poisson kernel rows alternate between $\coth(\pi y)$ and $\tanh(\pi y)$; the power terms are shift-independent and the tail bounds uniform in the shift) — locks $T(\Lambda_2)$, $T(\Lambda_1)$, and the final identity with half-widths 3.48×10^{-53} , 6.94×10^{-52} , and 9.12×10^{-53} . Direct torus integration at the complex parameter $c = 1.5128932208(1 + i)$ (outside Ast: functional 2.6357) confirms at float64 level: $n_4(3656 - 2600\sqrt{2}) = 3.528392069575363\dots$ versus $\text{EK}_4(\tau_4) = 3.528392069575611\dots$ (float64 noise floor).

8 The boundary of the method

We collect the precise limits of the machine, all visible already in Table 1.

8.1 Two-dimensional critical images

The decisive invariant is the dimension of the critical image (Remark 2.4). In the n_2 family the image is the slit $[0, 64]$, which does not separate $\mathbb{C}$, and every conjectured entry was reachable: the family is complete [5, 10, 11]. In the n_4 family the image is the astroid disc (4), which has interior. A CM parameter with $c(\tau_0)$ in the interior of Ast lies inside an open bad region of the τ -plane (open mapping), no certified path can approach it, and Theorem 2.10 is silent. The extreme case is the refuted conjecture $n_4(81) = 40M_7$ [6]: $c = 3$ is an interior lattice point of the astroid, the conjectured value equals the exact EK evaluation at the attached CM point on a *wrong sheet*, and direct torus integration gives the different value $n_4(81) = 4.1655349907533676508(5)$. Entries #4 and #5 of Table 1 have $c(\tau_0)$ inside Ast ($s \approx -2.45$ and $s \approx -6.17$ respectively) and are blocked by the same mechanism; entry #5 is moreover exterior (see the next subsection).

8.2 The wrong sheet below $\text{Im}\tau = 1/\sqrt{2}$

Even where the critical image is avoided, the EK series itself can fail. A systematic scan (`diag_n4_astroid.py`) shows $\text{EK}_4(\tau) \neq 4m(P + c(\tau))$ *everywhere* below the line $\text{Im}\tau = 1/\sqrt{2}$ — including points with $c(\tau)$ real and larger than 4, where the ${}_5F_4$ evaluation converges and the Mahler side is perfectly smooth (e.g. a difference of 11.4 at $\tau = 0.3i$). The difference grows continuously from 0 at the line; there is no jump. The mechanism is the same wrong-sheet phenomenon analyzed for the n_2 family in [5, Remark 5.6]: the single-valued series EK_4 follows a companion branch of $\tilde{m}$ once the branch locus is crossed. The propagation of Theorem 2.10 cannot cross, because the identity theorem fixes $\Psi \equiv 0$ only on the anchor's component W : along the imaginary axis the point $\tau = i/\sqrt{2}$ (where $c = 4$, a cusp of Ast) separates W from the lower component W' , and on W' the difference $H = F - G$ is non-constant. Entries #3, #9, and #10 of Table 1 (pure imaginary, $y = 0.612, 0.154$, and 0.463 , with $c > 4$ real in each case) are blocked at this *series* level: the conjectured identity — a statement about the true Mahler measure, which Samart's PSLQ

evidence supports — has no valid series input for station S5, and proving it requires a corrected expression, e.g. a modified Mahler measure of Samart–Tao $\tilde{n}$ -type [7], whose design principle (choose the linear combination so that the non-closed Deninger path [18] becomes a meaningful pairing) is compatible with stations S2/S4 of the machine.

8.3 Open directions

- *The ready entries* #1, #2, #6, #7, #8 of Table 1. For #1 and #7 the continuation step is already certified (Section 7); what remains is mechanical: `cert0` pair locks in discriminants -15 and -21 (class number 2; the genus machinery of Section 6 applies) and three-track P1 assemblies at levels 15, 84, and 24, 16.
- *The blocked entries* #3, #4, #5, #9, #10: a corrected (wrong-sheet-free) formula is needed before the machine has anything to certify; the $\tilde{n}$ framework of [7] is the natural candidate.
- *Samart’s three-modular- L -value hypothesis* $s = 16 + 1600\sqrt[3]{2} - 1280\sqrt[3]{4}$ [2]: the conjectured value involves three modular L -values and no attached CM point in the s_4 -image is known; the machine’s station S1 has no input.
- *The Boyd-lineage entries*: the two-variable conjectures of Boyd’s tables (e.g. conductor-14 and related families) have no known CM evaluation, so the machine does not apply; regulator-based approaches [18, 19], and in particular deconditioning of the currently conditional identities in the Brunault–Zudilin framework [19], remain the promising route.

Statements and Declarations

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Data Availability. All certification scripts, frozen certificates, and run logs are available at <https://github.com/huiminZheng-collab/samart-mahler>, archived at <https://doi.org/10.5281/zenodo.21711884>.

Use of AI tools. Documented in the Declaration on the use of AI tools on the title page.

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